

Aim

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Benchmarking: CITE-seq analysis

Code ▾

2026-08-05

Runs: pilot 260423_VH01624_453_222HWMYX for the first DSB evaluation and the CLR-vs-DSB comparison, then 260528_VH01624_464_222K7VKNX (LK2) for the production CITE QC.

Status: settled. DSB is the normalisation the project uses; CLR is kept only as the comparison baseline.

Analysis finished: 2026-07-07 - the most recent of the 42 figures shown below.

Report generated: 2026-08-05 by 07.cite_seq_report.Rmd, from the outputs of steps 01-06 in this folder.

ADT counts arrive from CITE-seq-Count as raw tag counts per barcode, and are dominated by ambient antibody. Two normalisations were compared: CLR, which scales within each cell, and DSB, which subtracts a background estimated from empty droplets and then removes per-cell technical noise using the isotype controls.

For full report, download cite_seq_report.html in the attachment and open in a browser.

Code used - attached and in the github repository:

https://github.com/nbartonicek/surface_proteome_dev/tree/main/lab_archives/5.Benchmarking-CITE-seq_analysis
(https://github.com/nbartonicek/surface_proteome_dev/tree/main/lab_archives/5.Benchmarking-CITE-seq_analysis)

Aim

Decide how to normalise the ADT signal, and confirm the counting step is not losing reads:

- does DSB give more specific markers than CLR
- is the background estimate stable, or an artefact of which isotype controls are used
- how many ADT reads come back unmapped, and whether the tag reference and read structure are correct

Conclusions

- **DSB gives increased marker specificity over CLR**, scored as mean of the expected population divided by median background, and is what the project uses
- the background estimate is stable: 30x random subsampling of the isotype controls does not move it materially
- populations are called at **3 SD over background**, at both the broad and the granular cell type level
- noise is estimated from ten isotype controls - IgG_Fc, Mouse IgG1, IgG2a and IgG2b, Rat IgG2b, IgG1_k, IgG1_l, IgG2a_k and IgG2c_k, and Armenian Hamster IgG
- mapping is not a problem: CITE-seq-Count maps **97% of ADT reads** and 95% of HTO reads
- the unmapped remainder is **poly-G** - 90 bases of G, 99.9% of the top-100 unmapped reads on the pilot ADT and 98.9% on HTO. That is the two-colour chemistry artefact, where a dark or failed cluster is base-called as G. These are not antibody tags, so they are neither a reference problem nor reads from discarded droplets
- step 06 as run is a whole-library check rather than an unmapped-read diagnosis, and the library passes it: the abundant sequences are exact matches to the reference in the correct orientation

1. Counting the tags

Step 01 runs CITE-seq-Count against the emptyDrops barcode whitelist, so the ADT matrix covers the same barcodes as the RNA. Step 02 is the same command with `-u` and `-ut 100000` added, which writes the unmapped tags out instead of discarding them - that table is what step 06 diagnoses. Both are kept because the production run does not produce the unmapped output, and the diagnostic run was done separately in July.

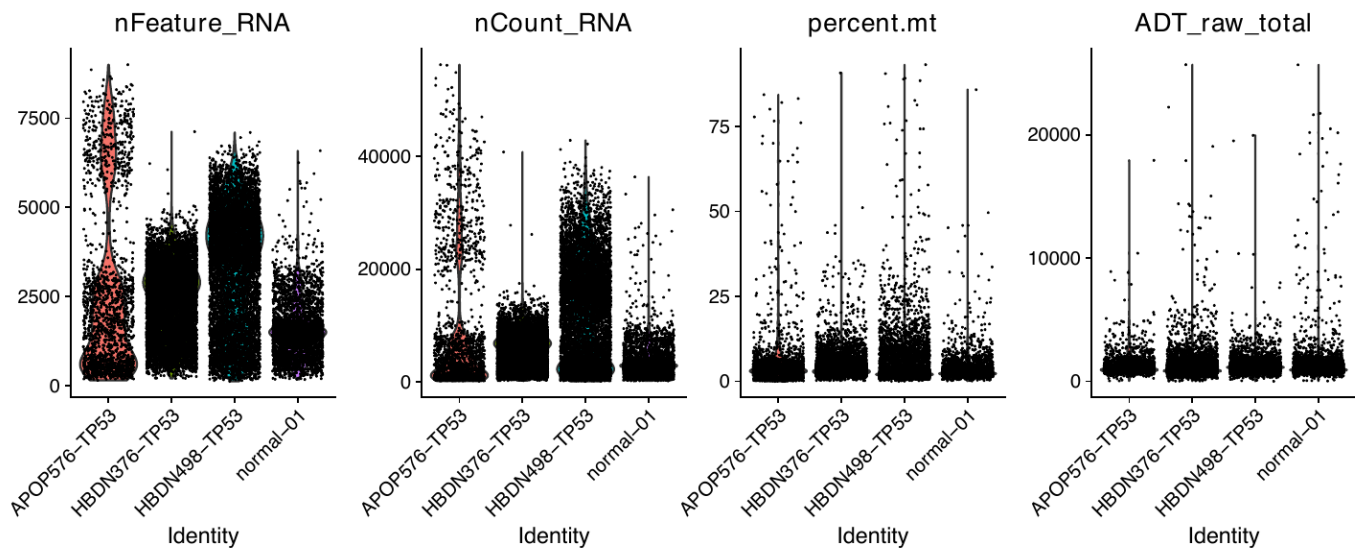
Note that the ADT is counted **outside** Cell Ranger: the Cell Ranger matrix for these runs contains Gene Expression features only. Anything that operates on the Cell Ranger matrix, CellBender included, therefore never sees the protein data.

2. CITE QC and DSB

Step 03 is the main workflow: load the demultiplexed object, import ADT counts for the emptyDrops cells, do the CLR-based QC first, then select background droplets from the raw output and run DSB with matched plots so the two can be compared like for like.

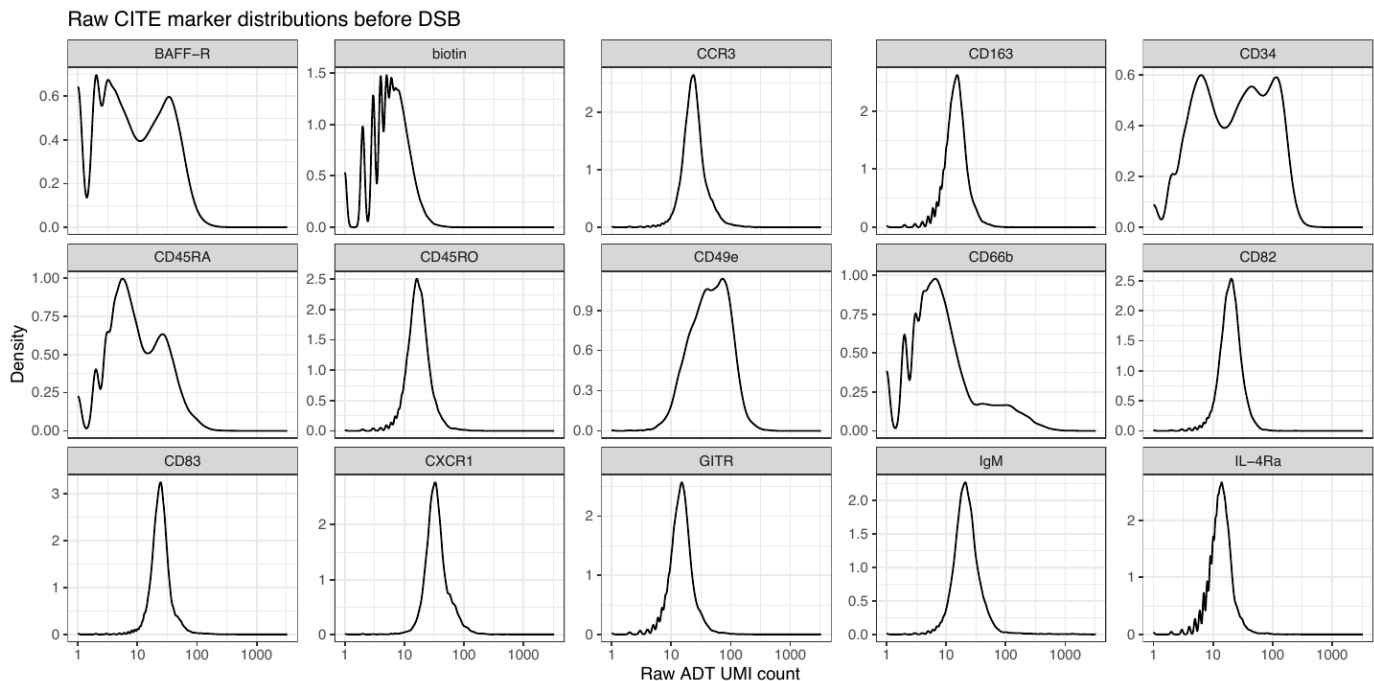
03_cell_QC_RNA_ADT_raw.pdf

Cell-level QC on RNA and raw ADT together.



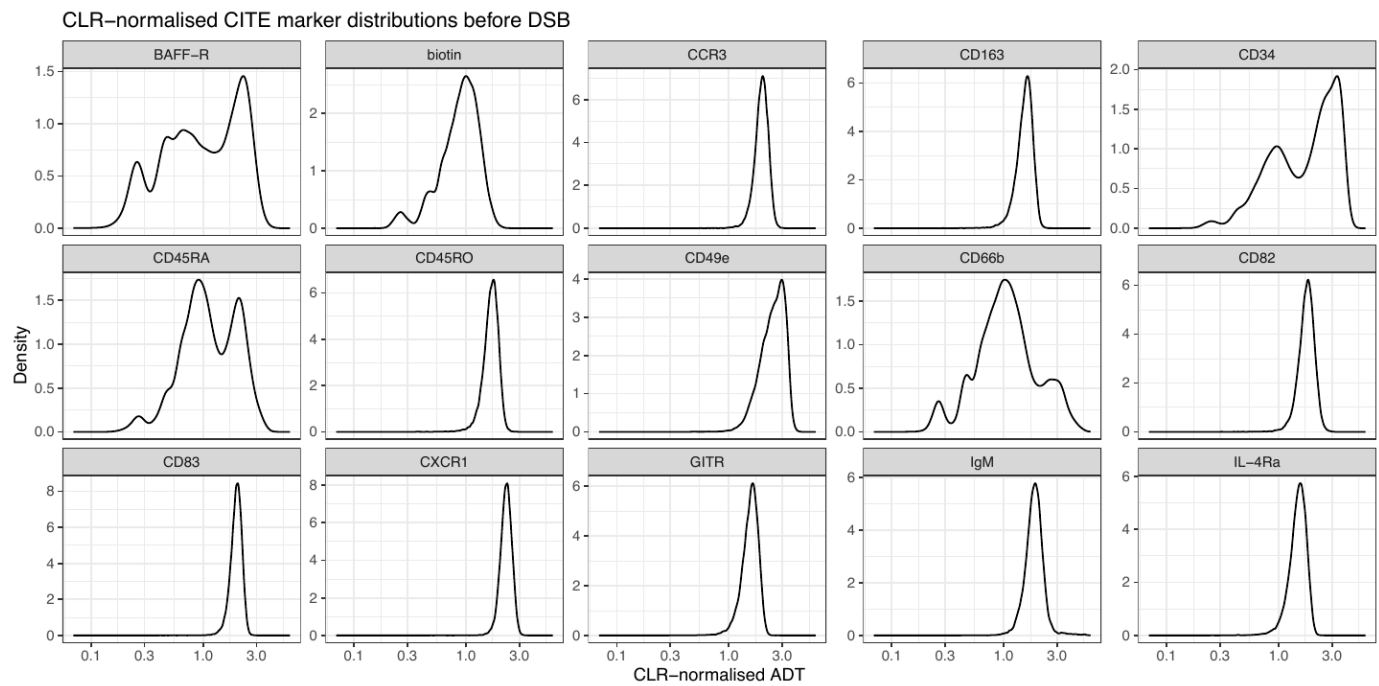
04_raw_ADT_marker_distributions.pdf

Raw ADT counts per marker, before any normalisation - the ambient signal is what makes these unusable as they stand.



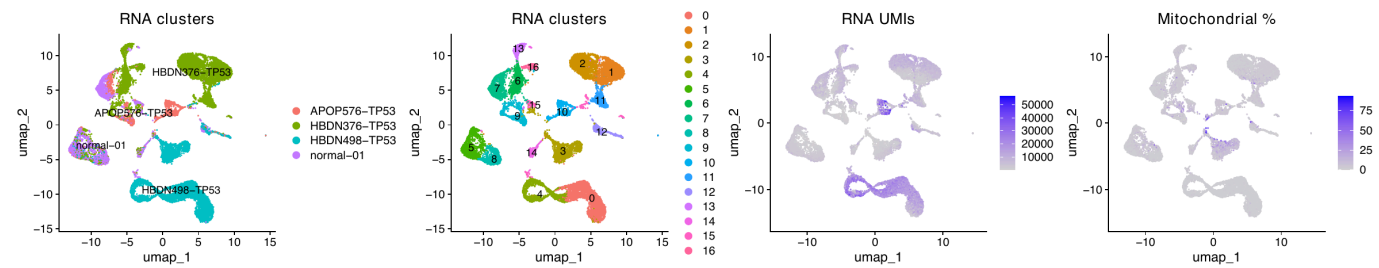
05_CLR_ADT_marker_distributions.pdf

The same markers after CLR.



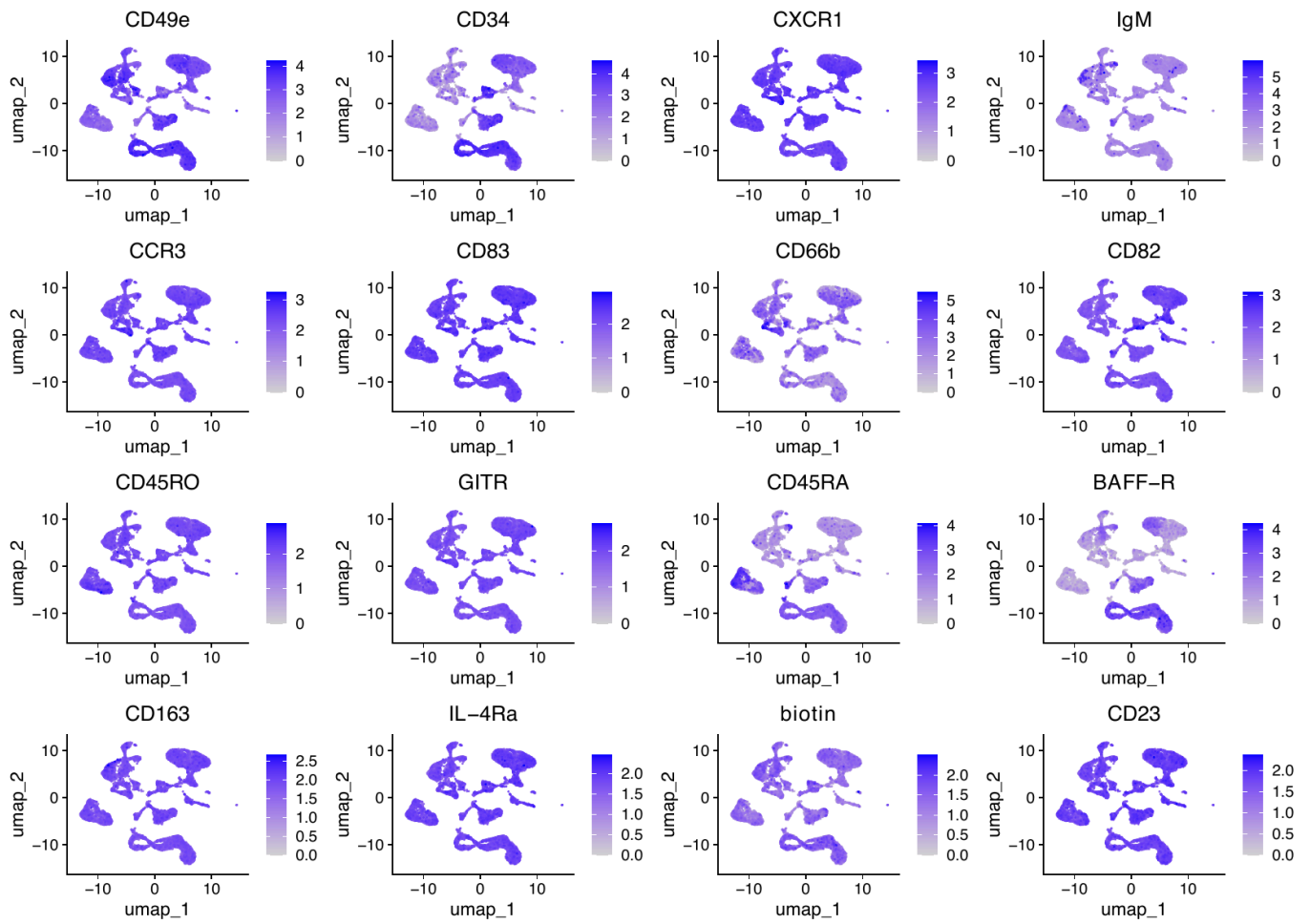
06_RNA_umap_QC.pdf

RNA UMAP used for the overlays below.



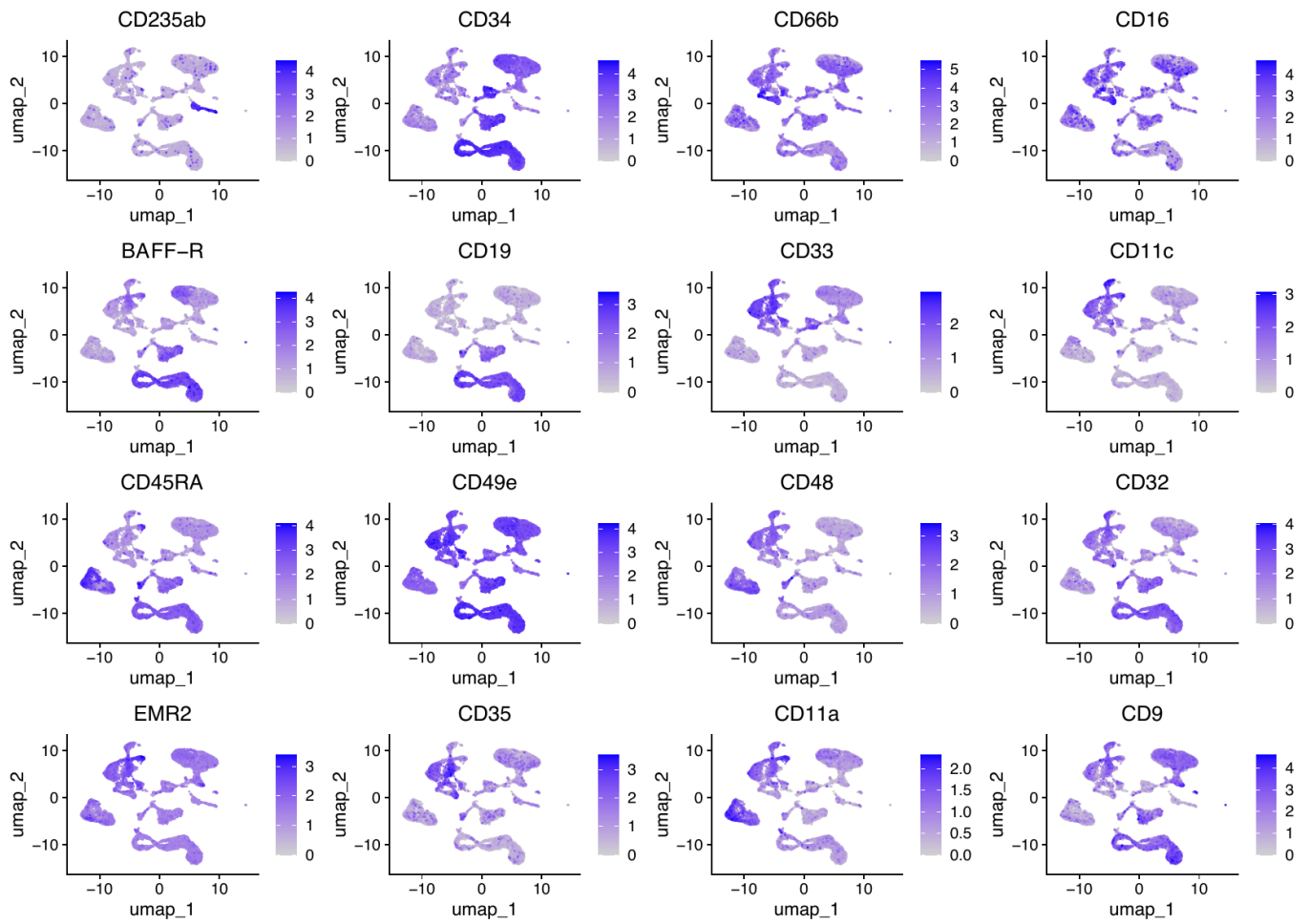
07_CLR_ADT_featureplots_top_markers.pdf

Top markers under CLR on the RNA UMAP.



08_CLR_ADT_discriminative_markers.pdf

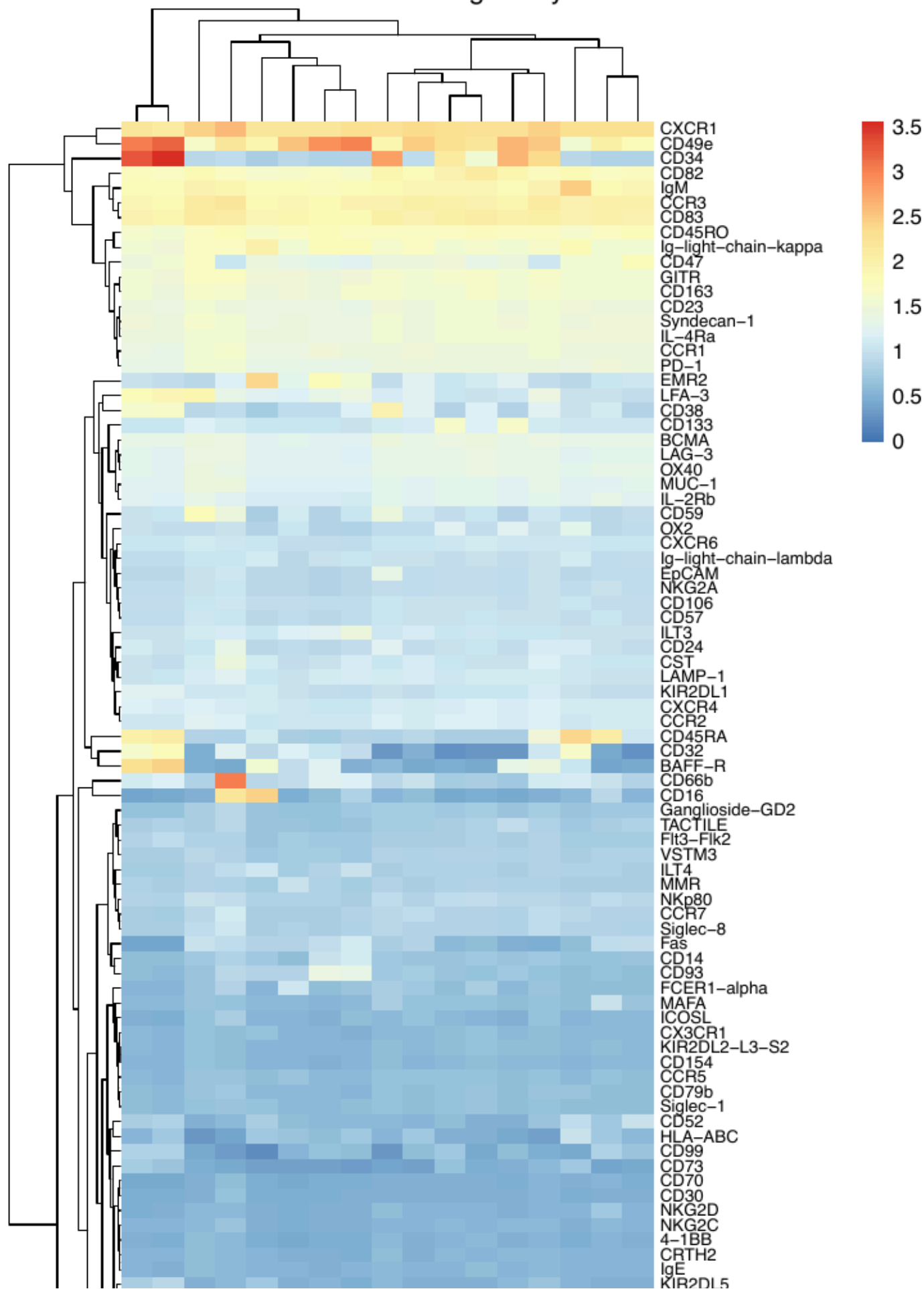
The most discriminative markers under CLR.

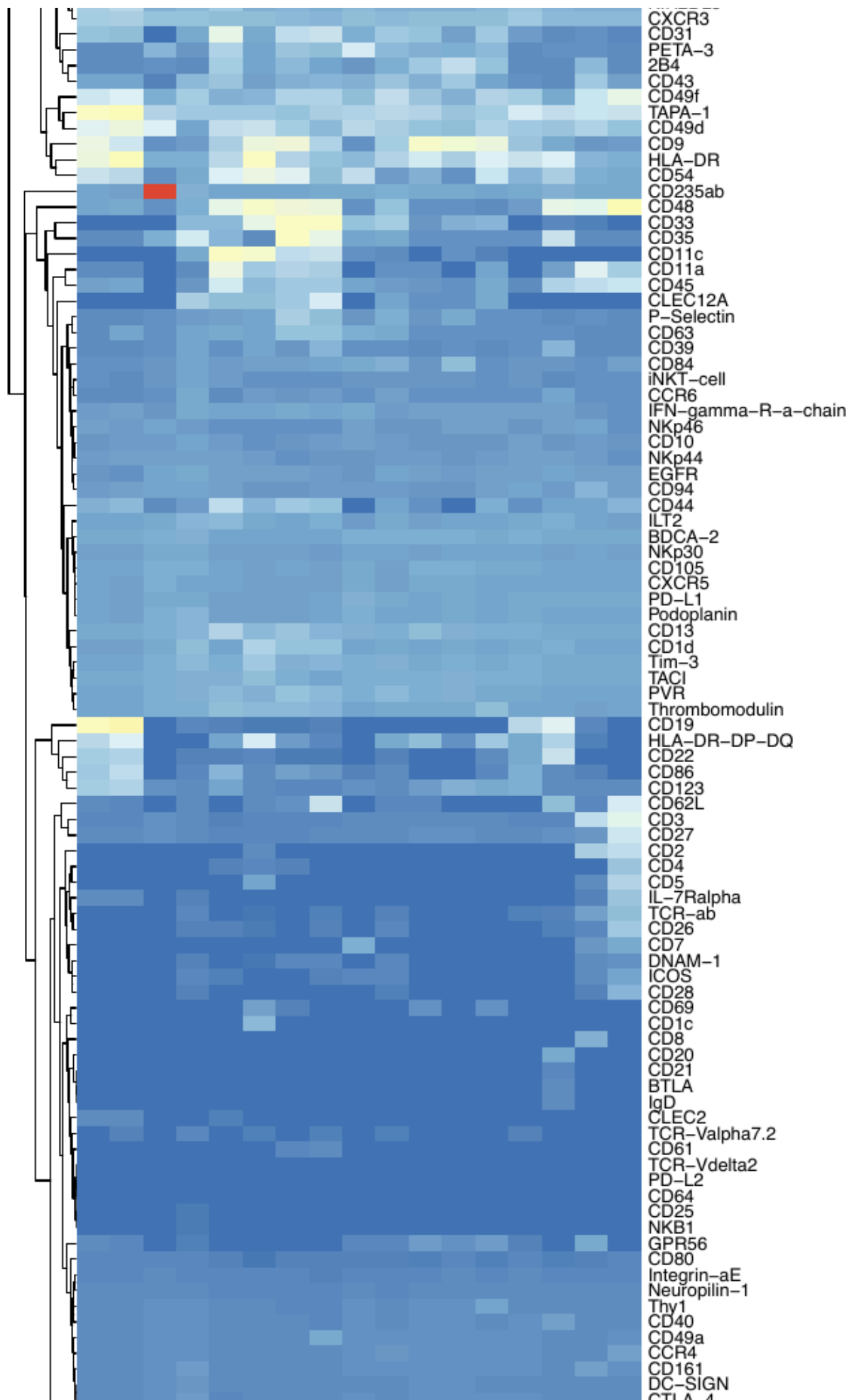


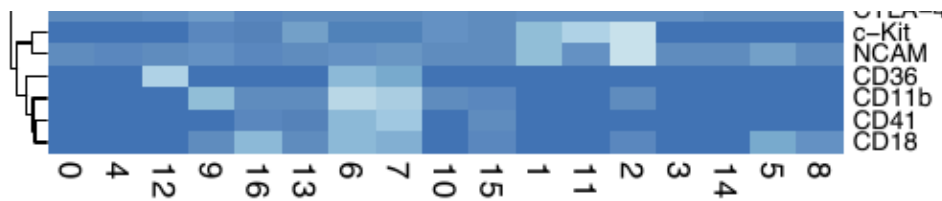
09_CLR_heatmap_by_RNA_cluster.pdf

CLR signal by RNA cluster.

Median CLR-normalised CITE signal by RNA cluster

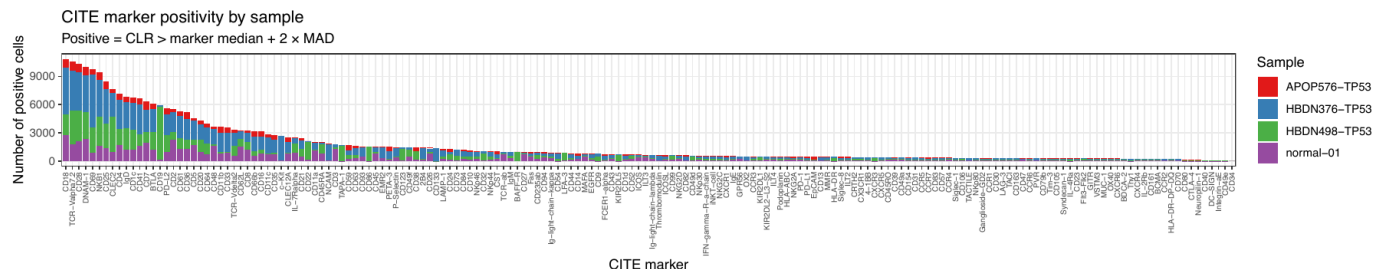






11_CITE_marker_CLR_positive_cells_by_sample.pdf

Positive cells per marker per sample under CLR.



3. Background and the isotype controls

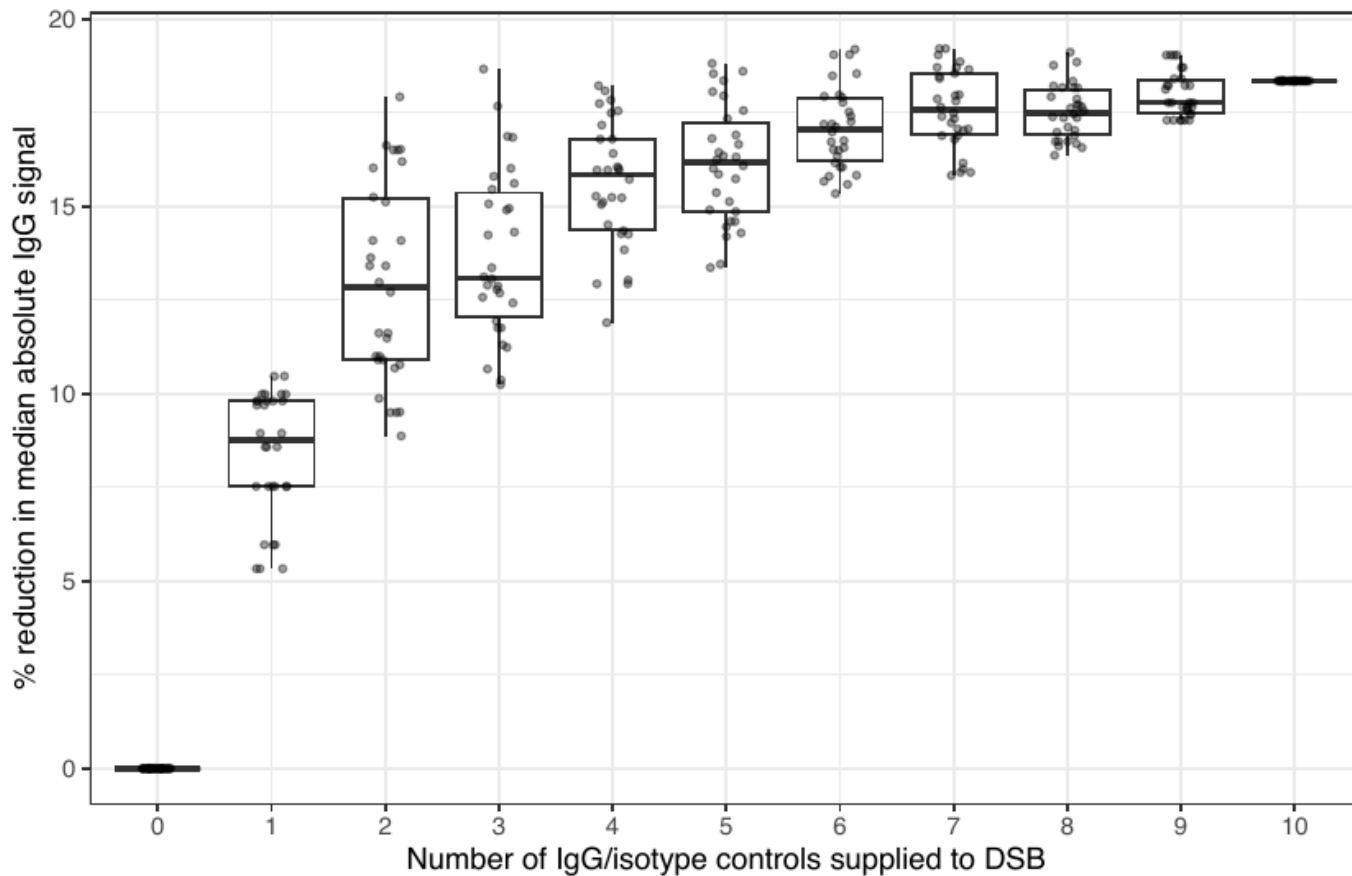
DSB estimates background from empty droplets and then removes per-cell technical noise using the isotype controls. Step 04 checks that estimate is not an artefact of which isotypes happen to be in the panel, by subsampling them 30 times at random and re-deriving the background.

dsb_igg_reduction_subsampling.pdf

Background estimate across 30 random subsamples of the isotype controls. A tight spread means the estimate does not depend on which controls are included, which is what justifies using it per cell.

DSB reduction of IgG/isotype control signal

Patient subset: HBDN206-MNpCT; relative to ambient-only DSB without isotype denoising

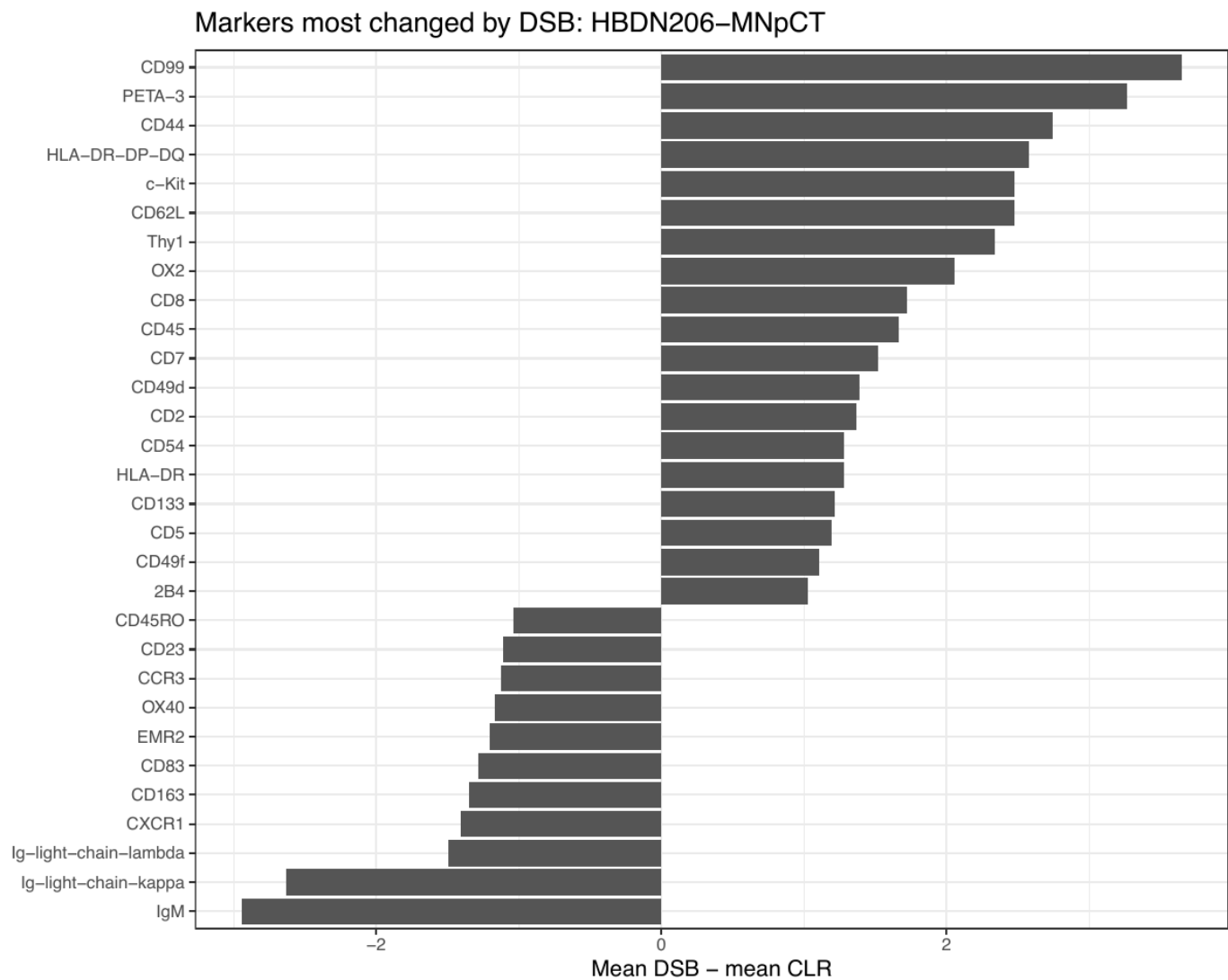


4. DSB against CLR

Step 05 is the head-to-head, on HBDN206-MNpCT.

01_marker_mean_shift_DSB_minus_CLR.pdf

Per-marker mean shift, DSB minus CLR. Markers above zero gained signal relative to background under DSB.



01_marker_mean_shift_DSB_minus_CLR.csv

marker	mean_CLR	mean_DSB	delta_DSB_minus_CLR
CD99	2.4733357	6.121970	3.6486344
PETA-3	1.7992250	5.067465	3.2682404
CD44	1.1138270	3.859165	2.7453384
HLA-DR-DP-DQ	1.8277369	4.404731	2.5769940
c-Kit	1.6256641	4.105634	2.4799703
CD62L	0.4517879	2.930737	2.4789489
Thy1	0.9490599	3.285325	2.3362655
OX2	1.9921708	4.052448	2.0602776
CD8	0.1453581	1.865855	1.7204967
CD45	1.1898704	2.858192	1.6683214
CD7	0.2110062	1.730692	1.5196859

marker	mean_CLR	mean_DSB	delta_DSB_minus_CLR
CD49d	1.5273281	2.914218	1.3868901
CD2	0.2679553	1.638304	1.3703489
CD54	1.1613030	2.443034	1.2817307
HLA-DR	2.3262512	3.603977	1.2777256
CD133	1.8782080	3.090959	1.2127506
CD5	0.1962089	1.391183	1.1949738
CD49f	1.0096267	2.116237	1.1066101
2B4	1.4927637	2.517694	1.0249307
CD123	1.0154373	1.879796	0.8643584
CD43	1.1120502	1.948497	0.8364467
CD31	0.8965285	1.686168	0.7896394
PVR	0.9623402	1.745876	0.7835355
CD10	0.4660673	1.240779	0.7747115
CD9	1.7797975	2.527111	0.7473134

02_RNA_protein_correlation_DSB_vs_CLR.csv

marker	gene	cor_CLR	cor_DSB	delta_DSB_minus_CLR
CD83	CD8A	-0.1190512	-0.0354092	0.0836420
CD82	CD8A	-0.1981759	-0.1319100	0.0662658
CD33	CD3D	-0.0941080	-0.0503718	0.0437362
CD31	CD3D	-0.1522775	-0.1096032	0.0426744
CD38	CD3D	0.0701710	0.1116796	0.0415085
CD163	FCGR3A	-0.0408244	-0.0094500	0.0313744
CD30	CD3D	-0.0314646	-0.0059678	0.0254968
CD45RA	PTPRC	0.0409043	0.0654681	0.0245638
CD45RO	PTPRC	0.1257468	0.1462185	0.0204717
CCR7	CCR7	-0.0015805	0.0176101	0.0191906
OX40	TNFRSF4	0.0331584	0.0506588	0.0175005
CD34	CD3D	-0.2802171	-0.2627862	0.0174309
CXCR1	CXCR1	0.0291581	0.0449373	0.0157792
CD32	CD3D	-0.0053044	0.0101965	0.0155010
CD83	CD83	0.0018763	0.0138721	0.0119958
CD39	CD3D	0.0167410	0.0279230	0.0111820
CD35	CD3D	-0.0161583	-0.0050585	0.0110998
CD49e	CD4	0.0713485	0.0822681	0.0109196
CD48	CD4	0.0072876	0.0177751	0.0104875
CD45RO	CD4	0.0107044	0.0199771	0.0092727
CD49a	CD4	0.0029852	0.0120500	0.0090647
IL-2Rb	IL2RB	0.1289432	0.1375641	0.0086208
CD4	CD4	0.0869039	0.0948451	0.0079412
CD41	CD4	0.0225840	0.0297711	0.0071870

marker	gene	cor_CLR	cor_DSB	delta_DSB_minus_CLR
CD47	CD47	0.0245270	0.0313452	0.0068182

The specificity score behind the conclusion is **mean of the expected population divided by median background**, computed per marker against a hand-curated panel: CD163, CD14, CD33 and CD11b for myeloid/DC; CD34, CD117 and CD133 for stem/progenitor; CD4, CD8, CD19 and CD20 for lymphoid; CD71 for erythroid. Cells are called positive at 3 SD over background.

5. Unmapped ADT reads

Read this section for what it is. Step 06 was run with `input_mode <- "full_fastq"` against the entire LK2 ADT R2 file, not against the unmapped reads. Everything below therefore describes the whole library, and is a check that the tag reference and read structure are right - which they are. It is not a diagnosis of the unmapped reads.

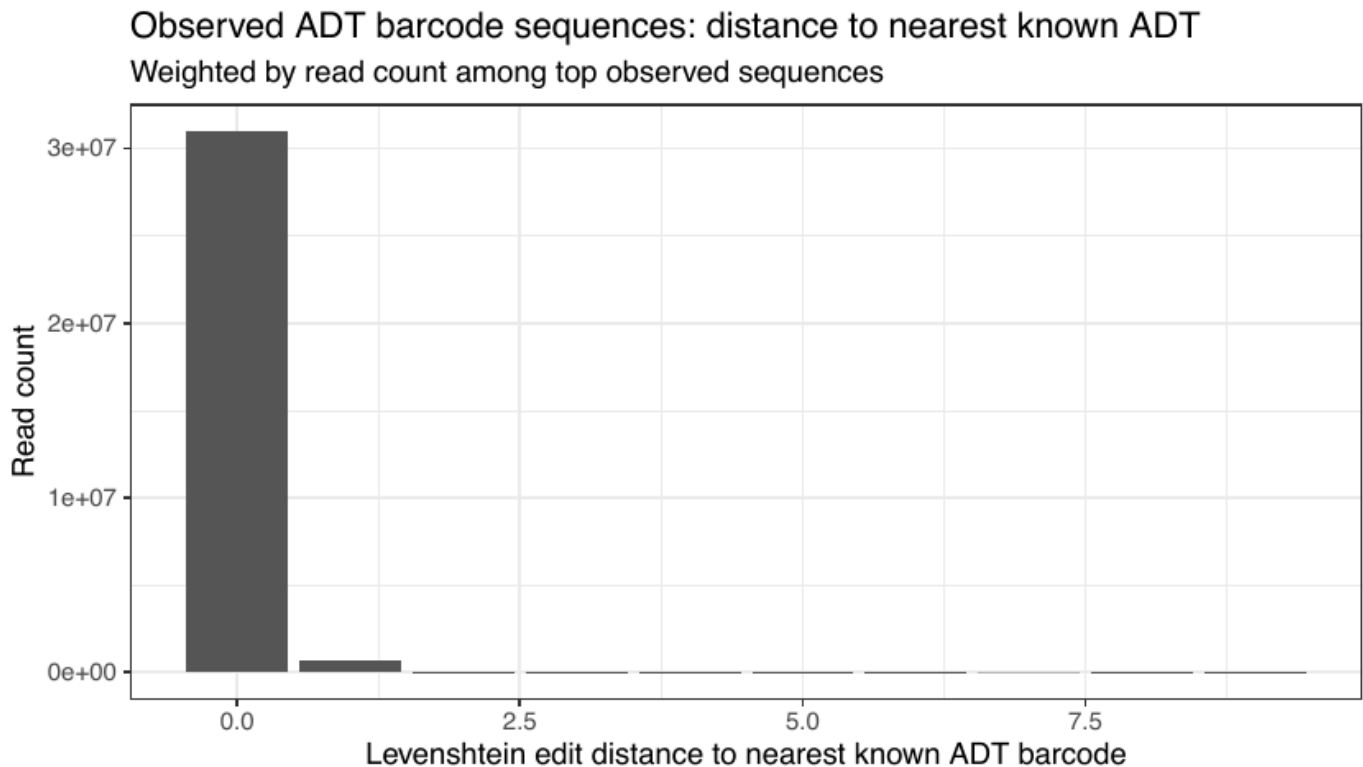
The unmapped reads are separately accounted for. CITE-seq-Count's own `run_report.yaml` puts mapping at **97%** for ADT and **95%** for HTO, and its `unmapped.csv` shows what the remainder is:

Run / library	Poly-G share of the top-100 unmapped
Pilot ADT (260423 , LK1)	99.9%
Pilot HTO (260423 , LK1)	98.9%
LK2 ADT (260528)	67.5%

Those sequences are 90 bases of G . On two-colour chemistry a dark or failed cluster is base-called as G, so this is empty flowcell rather than anything about the antibodies, the reference or the droplets. `unmapped.csv` lists only the top 100 sequences, so this characterises the dominant species rather than every unmapped read - and the LK2 figure of 67.5% means a third of its unmapped reads are something else, which has not been looked at.

[edit_distance_histogram.pdf](#)

Edit distance from each observed barcode to the nearest known tag. A pile-up at distance 1-2 means sequencing error; a pile-up further out means the observed sequences are not the tags at all.

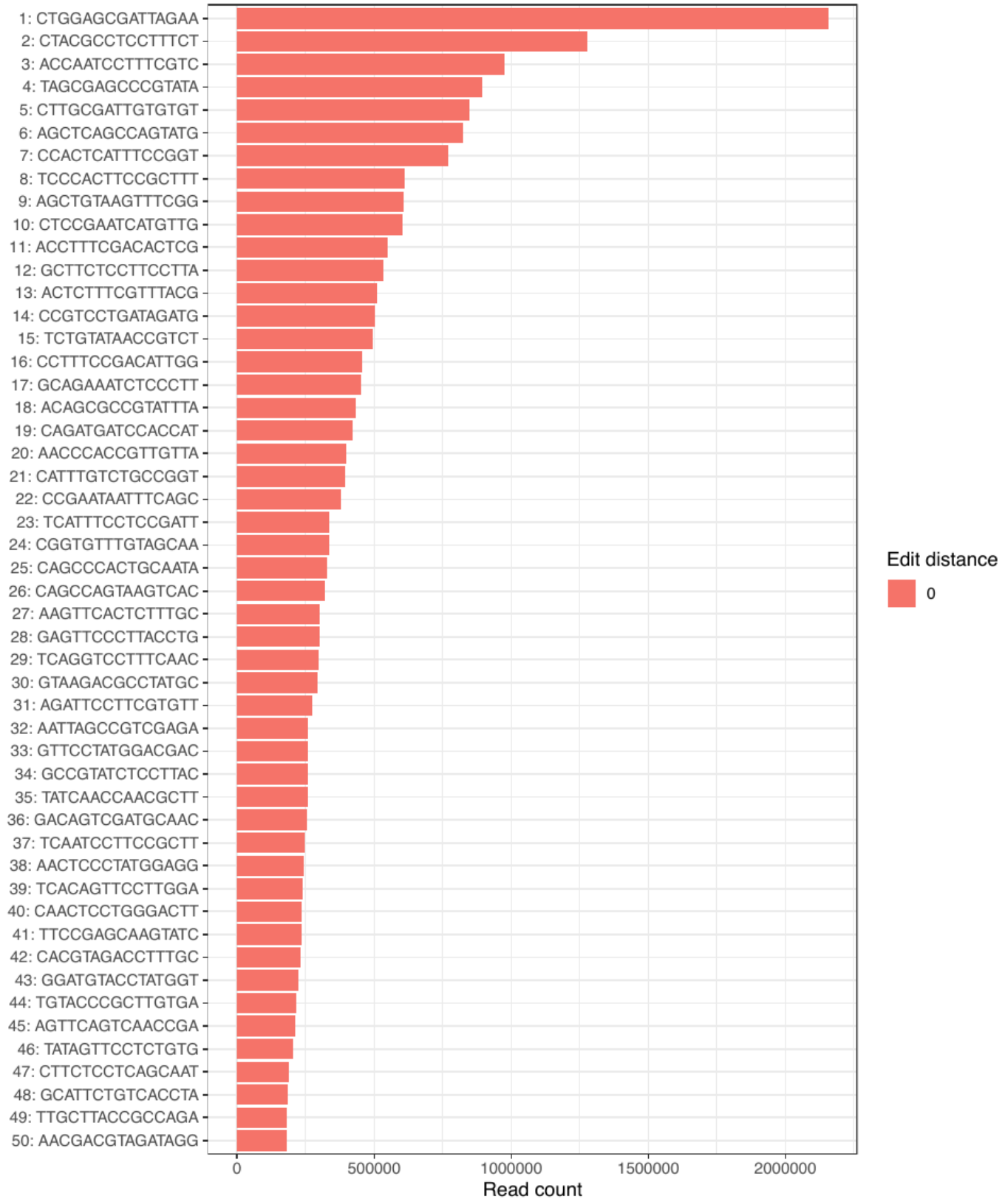


[top_observed_barcodes.pdf](#)

The most frequent observed barcodes among the unmapped reads.

Top observed ADT barcode sequences

Fill = edit distance to nearest known barcode



observed_barcode_counts.csv

observed_sequence	read_count	rank	pct_reads
CTGGAGCGATTAGAA	2155592	1	6.492747
CTACGCCTCCTTTCT	1278629	2	3.851292

observed_sequence	read_count	rank	pct_reads
ACCAATCCTTTCGTC	973419	3	2.931985
TAGCGAGCCCGTATA	894278	4	2.693608
CTTGCGATTGTGTGT	848714	5	2.556367
AGCTCAGCCAGTATG	825766	6	2.487247
CCACTCATTTCCGGT	768091	7	2.313527
TCCCACTTCCGCTTT	611319	8	1.841322
AGCTGTAAGTTTCGG	607190	9	1.828885
CTCCGAATCATGTTG	602057	10	1.813425
ACCTTTCGACACTCG	547159	11	1.648069
GCTTCTCCTTCCTTA	532205	12	1.603027
ACTCTTTCGTTTACG	509711	13	1.535274
CCGTCCTGATAGATG	502076	14	1.512277
TCTGTATAACCGTCT	493087	15	1.485202
CCTTTCGACATTGG	455018	16	1.370536
GCAGAAATCTCCCTT	451726	17	1.360621
ACAGCGCCGTATTTA	434361	18	1.308316
CAGATGATCCACCAT	420574	19	1.266789
AACCCACCGTTGTTA	398800	20	1.201205

observed_vs_known_distance.csv

observed_sequence	read_count	rank	pct_reads	nearest_known_sequence	nearest_antibody	edit_distance	exact_match	near_match
CTGGAGCGATTAGAA	2155592	1	6.492747	CTGGAGCGATTAGAA	IgG_Fc	0	TRUE	TRUE
CTACGCCTCCTTTCT	1278629	2	3.851292	CTACGCCTCCTTTCT	CXCR1	0	TRUE	TRUE
ACCAATCCTTTCGTC	973419	3	2.931985	ACCAATCCTTTCGTC	CCR3	0	TRUE	TRUE
TAGCGAGCCCGTATA	894278	4	2.693608	TAGCGAGCCCGTATA	IgM	0	TRUE	TRUE
CTTGCGATTGTGTGT	848714	5	2.556367	CTTGCGATTGTGTGT	CD49e	0	TRUE	TRUE
AGCTCAGCCAGTATG	825766	6	2.487247	AGCTCAGCCAGTATG	Ig_light_chain_kappa	0	TRUE	TRUE
CCACTCATTTCCGGT	768091	7	2.313527	CCACTCATTTCCGGT	CD83	0	TRUE	TRUE
TCCCACTTCCGCTTT	611319	8	1.841322	TCCCACTTCCGCTTT	CD82	0	TRUE	TRUE
AGCTGTAAGTTTCGG	607190	9	1.828885	AGCTGTAAGTTTCGG	CD66b	0	TRUE	TRUE
CTCCGAATCATGTTG	602057	10	1.813425	CTCCGAATCATGTTG	CD45RO	0	TRUE	TRUE
ACCTTTCGACACTCG	547159	11	1.648069	ACCTTTCGACACTCG	GITR	0	TRUE	TRUE
GCTTCTCCTTCCTTA	532205	12	1.603027	GCTTCTCCTTCCTTA	CD163	0	TRUE	TRUE
ACTCTTTCGTTTACG	509711	13	1.535274	ACTCTTTCGTTTACG	Syndecan_1	0	TRUE	TRUE
CCGTCCTGATAGATG	502076	14	1.512277	CCGTCCTGATAGATG	IL-4Ra	0	TRUE	TRUE
TCTGTATAACCGTCT	493087	15	1.485202	TCTGTATAACCGTCT	CD23	0	TRUE	TRUE
CCTTTCGACATTGG	455018	16	1.370536	CCTTTCGACATTGG	CCR1	0	TRUE	TRUE
GCAGAAATCTCCCTT	451726	17	1.360621	GCAGAAATCTCCCTT	CD34	0	TRUE	TRUE
ACAGCGCCGTATTTA	434361	18	1.308316	ACAGCGCCGTATTTA	PD-1	0	TRUE	TRUE
CAGATGATCCACCAT	420574	19	1.266789	CAGATGATCCACCAT	BCMA	0	TRUE	TRUE
AACCCACCGTTGTTA	398800	20	1.201205	AACCCACCGTTGTTA	OX40	0	TRUE	TRUE

reverse_complement_distance.csv

read_count	rank	pct_reads	observed_sequence	nearest_known_sequence	nearest_antibody	edit_distance	exact_match	near_match
2155592	1	6.492747	TTCTAATCGCTCCAG	ATCACATCGTTGCCA	PVR	6	FALSE	FALSE
1278629	2	3.851292	AGAAAGGAGGCGTAG	AGCTACGAGCAGTAG	HLA-DR_DP_DQ	5	FALSE	FALSE
973419	3	2.931985	GACGAAAGGATTGGT	GACAAGTGATCTGCA	CD11b	6	FALSE	FALSE
894278	4	2.693608	TATACGGGCTCGCTA	TATGCGAGGCTTATC	HLA-ABC	6	FALSE	FALSE
848714	5	2.556367	ACACACAATCGCAAG	GACAGTCGATGCAAC	CXCR6	6	FALSE	FALSE
825766	6	2.487247	CATACTGGCTGAGCT	CAACTCTGGGACTT	NKG2A	6	FALSE	FALSE
768091	7	2.313527	ACCGGAAATGAGTGG	ACGAATCAATCTGTG	CD80	7	FALSE	FALSE
611319	8	1.841322	AAAGCGGAAGTGGGA	AATAGCGAGCAAGTA	HLA-DR	6	FALSE	FALSE
607190	9	1.828885	CCGAAACTTACAGCT	CCGAATAATTTACAGC	MUC-1	4	FALSE	FALSE
602057	10	1.813425	CAACATGATTCGGAG	CAGTAAGTTCGGGAC	4-1BB	6	FALSE	FALSE
547159	11	1.648069	CGAGTGTGCGAAAGGT	GGATGTACCTATGGT	EMR2	6	FALSE	FALSE
532205	12	1.603027	TAAGGAAGGAGAAGC	AACGACGTAGATAGG	ILT4	6	FALSE	FALSE
509711	13	1.535274	CGTAAACGAAAGAGT	CGTAACGTAGAGCGA	TCR_ab	5	FALSE	FALSE
502076	14	1.512277	CATCTATCAGGACGG	CAGCCATTATTAGG	CD10	6	FALSE	FALSE
493087	15	1.485202	AGACGGTTATACAGA	AGACTAATAGCTGAC	c-Kit	6	FALSE	FALSE
455018	16	1.370536	CCAATGTCGGAAAGG	ACAATTTGAACAGCG	NKp46	6	FALSE	FALSE
451726	17	1.360621	AAGGGAGATTTCTGC	CACGTAGACCTTTGC	OX2	6	FALSE	FALSE
434361	18	1.308316	TAAATACGGCGCTGT	TAACTCAGGGCCTAT	CD33	5	FALSE	FALSE
420574	19	1.266789	ATGGTGGATCATCTG	ATCGTCGAGAGCTAG	CD105	6	FALSE	FALSE
398800	20	1.201205	TAACAACGGTGGGTT	TATCAACCAACGCTT	KIR2DL1	6	FALSE	FALSE

The four outputs test four explanations in turn: whether the observed sequences resemble the known tags at all, how far they are from them, whether the tag sits at a different offset in the read than expected, and whether it has been reverse complemented. Across the top 1,000 observed sequences, covering 96% of reads, 93 percentage points are exact matches to a real antibody and 96 are within two edits, while the reverse-complement check sits at distance 5-7 with no matches. The library and reference are sound.

To diagnose the unmapped reads properly, step 06 would be re-run with `input_mode <- "unmapped_fastq"` against the unmapped output from step 02 (`cite_seq_count/<run>/unmapped_tags_<sample>.csv`).

6. What was run, in order

Step	Script	What it does
01	01.cite_seq_count.sbatch	CITE-seq-Count on the ADT library, emptyDrops whitelist. Production counting.
02	02.cite_seq_count_unmapped.sbatch	Same run with <code>-u / -ut 100000</code> so unmapped tags are written out.
03	03.CITE_qc.R	CITE QC then DSB: raw totals, CLR, marker distributions, overlays, heatmaps, background selection, DSB.
04	04.evaluate_DSB.R	DSB against background droplets with 30x isotype subsampling.
05	05.compare_DSB_vs_CLR.R	Head-to-head DSB vs CLR on one sample.
06	06.CITE_unmapped_QC.R	Diagnoses the unmapped ADT reads.
07	07.cite_seq_report.Rmd	This report.

scripts/7d.DSB_cluster_resolution.R is not a separate step: it is step 05 re-pointed at LK2, differing by twelve lines that are all run and sample variables. Change run , sample_name and sample_short in step 05 to reproduce it. Its name is misleading - it is not a cluster-resolution analysis.

7. Data locations

What	Location
ADT counts	results/cite_seq_count/<run>/adt_counts_*/
CITE QC and DSB	results/cite_seq_dsb/<run>/
Earlier QC-then-DSB outputs	results/benchmarking/cite_qc_then_dsb/
Isotype subsampling	results/benchmarking/dsb_igg_subsample/
DSB vs CLR	results/seurat_annotated/<run>/projectionFigures/DSB_vs_CLR_<donor>/
Unmapped diagnostics	results/adt_unmapped_diagnostics/ and results/adt_unmapped_investigator/

8. Tools

Tool	Version	Source
CITE-seq-Count	to confirm	steps 01 and 02
dsb	see session below	denoised and scaled by background
Seurat CLR	see session below	comparison baseline

```
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.5.2
##
## Matrix products: default
## BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.1
##
## locale:
## [1] en_AU/en_AU/en_AU/C/en_AU/en_AU
##
## time zone: Australia/Melbourne
## tzcode source: internal
##
## attached base packages:
## [1] stats graphics grDevices utils datasets methods base
##
## other attached packages:
## [1] scales_1.4.0 ggplot2_4.0.2 knitr_1.51 purrr_1.2.1 stringr_1.6.0
## [6] readr_2.1.6 tidyr_1.3.2 dplyr_1.2.0
##
## loaded via a namespace (and not attached):
## [1] bit_4.6.0 gtable_0.3.6 jsonlite_2.0.0 crayon_1.5.3
## [5] qpdf_1.4.1 compiler_4.5.1 pdftools_3.9.0 Rcpp_1.1.1
## [9] tidyselect_1.2.1 parallel_4.5.1 jquerylib_0.1.4 yaml_2.3.12
## [13] fastmap_1.2.0 R6_2.6.1 generics_0.1.4 tibble_3.3.1
## [17] RColorBrewer_1.1-3 bslib_0.10.0 pillar_1.11.1 tzdb_0.5.0
## [21] rlang_1.1.7 cachem_1.1.0 stringi_1.8.7 xfun_0.56
## [25] S7_0.2.1 sass_0.4.10 bit64_4.6.0-1 ote_0.2.0
## [29] cli_3.6.5 withr_3.0.2 magrittr_2.0.4 digest_0.6.39
## [33] grid_4.5.1 vroom_1.7.0 askpass_1.2.1 hms_1.1.4
## [37] lifecycle_1.0.5 vctrs_0.7.1 evaluate_1.0.5 glue_1.8.0
## [41] farver_2.1.2 rmarkdown_2.30 tools_4.5.1 pkgconfig_2.0.3
## [45] htmltools_0.5.9
```

9. Caveats

1. **The residual CITE analysis is not included.** `scripts/13.CITE_residuals.R` belongs to this period but cannot be run as written - it never loads an object, expecting `seu_dsb` and `celltype_cols` to exist in the session already - and its output directory `results/benchmarking/residual_cite_analysis` is empty. It will be added once it has been run.
2. **The CLR-vs-DSB comparison is one donor**, HBDN206-MNpCT on the pilot run. The LK2 equivalent exists only as the re-pointed script noted above.
3. **CellBender does not touch the protein data** in the current pipeline, since the Cell Ranger matrix is `Gene Expression` only. Whether ambient protein removal before DSB helps is a separate question, and is not tested here.